CHAPTER

4

DSP Software

DSP applications are usually programmed in the same languages as other science and engineering tasks, such as: C, BASIC and assembly. The power and versatility of C makes it the language of choice for computer scientists and other professional programmers. On the other hand, the simplicity of BASIC makes it ideal for scientists and engineers who only occasionally visit the programming world. Regardless of the language you use, most of the important DSP software issues are buried far below in the realm of whirling ones and zeros. This includes such topics as: how numbers are represented by bit patterns, round-off error in computer arithmetic, the computational speed of different types of processors, etc. This chapter is about the things you can do at the *high level* to avoid being trampled by the *low level* internal workings of your computer.

Computer Numbers

Digital computers are very proficient at storing and recalling numbers; unfortunately, this process isn't without error. For example, you instruct your computer to store the number: 1.41421356. The computer does its best, storing the closest number it *can* represent: 1.41421354. In some cases this error is quite insignificant, while in other cases it is disastrous. As another illustration, a classic computational error results from the addition of two numbers with very different values, for example, 1 and 0.00000001. We would like the answer to be 1.00000001, but the computer replies with 1. An understanding of how computers store and manipulate numbers allows you to anticipate and correct these problems *before* your program spits out meaningless data.

These problems arise because a fixed number of bits are allocated to store each number, usually 8, 16, 32 or 64. For example, consider the case where eight bits are used to store the value of a variable. Since there are $2^8 = 256$ possible bit patterns, the variable can only take on 256 different values. This is a fundamental limitation of the situation, and there is nothing we can do about it. The part we *can* control is what value we declare each bit pattern